



The Oracle Asymmetry

Where evolutionary self-improvement of an agent gets measurable traction

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Organization: Agent Self-Evolution

Repository: github.com/jramos/agent-self-evolution

Abstract

A framework for evolving an agent's skills, tool descriptions, prompts, and code was exercised across many experiments. The consolidated result is a **spend-allocation** finding shaped like an asymmetry. Self-evolution produced deploy-grade traction under a **conjunction** of conditions — a ground-truth executable oracle, real headroom, and a re-derivation task. On real historical bugs this re-derived a correct, oracle-matching fix for **12 of 20** bugs (deploy-reachable **0.60**, Wilson 95% CI [0.39, 0.78]; it clears a pre-registered **futility floor** of 0.10, which certifies "not catastrophic," not a validated deployment rate). Where the fitness signal is instead an LLM judge or a capable agent's behavior, we measured **no detectable effect** — but at a power that resolves only *large* couplings (see below). The two arms are instruments of unequal resolution, and oracle-presence is confounded with headroom and task type, so this is a direction-setting result — not a proof of inertness, and not a single "oracle vs. agent" law.

1. The positive result: executable-oracle code re-derivation

Real historical tool bugs were harvested from an active repository's git stream, each carrying its upstream fix commit as a **ground-truth oracle**. The loop repairs the buggy parent in an isolated worktree from failing-test feedback; a repair counts only if it passes the full fix-commit test set and behaviorally matches the upstream fix. The organism (one bug) is the unit, and a bug is deploy-reachable when a majority of 3 independent seeds produce an oracle-matching fix.

Estimand (organism-level)	Value
Deploy-reachable	12 / 20 = 0.60
Wilson 95% CI	[0.387, 0.781]
Cluster bootstrap (mean, CI)	0.602, [0.40, 0.80]
Clears futility floor (0.10)?	yes - P(below) = 0.0
ICC / design effect / effective-N	0.326 / 1.65 / 36.3
Cost	\$3.10

Read the verdict honestly. The 0.10 line is a futility floor, not a success bar — deploy-reachable would have to collapse to 4/20 before it failed. The honest headline is the point estimate **0.60 [0.39, 0.78] on one repository, one proposer model**: promising, not settled. The number is also fragile at the knee: the per-seed distribution is bimodal (0 correct: 1 organism, 1: 7, 2: 2, 3: 10), so 10 organisms are unanimous successes, 8 are clear failures, and **0.60 hinges on the 2 organisms sitting exactly at 2/3** — each one seed-flip from flipping the verdict. The cluster bootstrap resamples the already-dichotomized flags, so [0.40, 0.80] does not propagate the 3-seed noise on each organism; a two-stage bootstrap would be wider. The pooled per-seed rate (41/60 = 0.68) is recorded only for contrast and labelled dishonest in the source artifact (ICC 0.33 inflates its precision); 0.60 is a different, threshold-dependent estimand, not a de-biased 0.68.

Validity threat (not closed): the repair loop feeds the proposer the failing test's full output, which for assertion failures includes the expected values, and the measurement gate verifies an oracle match with the held-out split off — so 0.60 mixes genuine re-derivation with possible teach-to-the-test transcription (and possible public-repo recall). A targeted check — re-running with traceback feedback suppressed, and inspecting accepted rewrites for hard-coded literals — is the right next step before treating 0.60 as a pure capability rate. Of 33 harvested candidates: 20 valid organisms, 7 source-missing, 5 not-valid, 1 too-large.

2. The null: capable-agent artifact decoupling

The sharp, defensible form of this result is about **inferability**, not behavior in general: when a capable agent (gpt-5-mini class) can recover a tool's behavior from its **name plus the task**, the description text is redundant, and editing it — better or worse — does not move behavior.

- **Improving did not help.** Evolving skills, tool descriptions, and turn-level sequences produced null deploy results, including under a judge-free test-command verdict (so judge saturation is not the cause). These baselines were also largely already-saturated / hand-tuned — low headroom, so little to gain.
- **Degrading did not hurt.** A strongly misdirecting description, and even an actively false one ("merges and preserves content, never deletes"), each produced 0/7 caught-losses against a 0/7 A/A false-alarm floor. The probe tool's name states its function — the weakest possible test of description sensitivity, which is exactly why the agent routes past the text.

Power is the binding caveat. With $n=7$ per arm, 0/7-vs-0/7 resolves only large couplings: a one-sided Fisher test cannot separate treatment from the floor until the caught-loss rate reaches about 4/7 (~57%); at 80% power the minimum detectable effect is roughly 40%. So "no detectable effect" means we can rule out only effects larger than about half — not "we had good power and saw nothing." A delivery canary (a description that changes which file is written) separates 1.00 vs 0.00, proving the harness records routing; it does not prove the harness could detect a realistic, intermediate-magnitude quality effect — that control was not run. This is a failure to reject the null on one capable-agent class, for name-inferable tools, on low-headroom artifacts; it bounds the effect and predicts the coupling reappears where the agent cannot infer behavior from priors.

3. The dead ends: two honestly-distinct kinds

Supply-absent (a real boundary). Over a year of one active repo's history there were zero clean, tool-local dependency-version regressions: manifest-touching commits were broad migrations or feature additions. This delimits regardless of effect size — the instances do not exist in the corpus.

Underpowered nulls (not boundaries). These found no usable signal at small N , but their intervals are wide — they steer spend, they do not prove impossibility:

- **Metamorphic verification (no oracle): 0/8.** Generic invariants caught 0 of 8 real bugs (Wilson upper ~ 0.37 , consistent with metamorphic working $\sim 1\text{-in-}4$). Real tool bugs are input-specific edge cases that do not violate broad invariants.
- **Held-out-split independence: 3/8.** Only 3 of 8 bugs had tests input-diverse enough for an independent held-out, so the independent held-out product is deferred — the oracle sidesteps it for measurement.

4. The unifying thesis, and its limit

Across the classes, traction tracked **how mechanical and agent-free the verdict is**: a deterministic test on executed code yielded a gradient; a judge or a capable agent's behavior did not. But the contrast is between two instruments of unequal resolution (one resolves effects of 0.10, the other only ~ 0.5), and oracle-presence is confounded with headroom and task type, so the clean "oracle vs. agent" axis is a hypothesis this campaign suggests but did not isolate. The disconfirming experiment never run: apply the executable-oracle loop to an already-correct function (oracle present, no headroom) — if the gradient vanishes, headroom, not the oracle, was load-bearing. Practically, the implication stands: spend evolutionary budget where an executable oracle and real headroom exist (code, fed by the shipped propose-only triage sentinel), and not on artifact quality for capable, name-inferable surfaces.

5. Limitations

- **Teach-to-the-test (top threat).** The proposer sees expected values in test tracebacks and the held-out split is off, so 0.60 may overstate genuine re-derivation until the leakage check is run.
- **Underpowered null.** $n=7$ resolves only effects above $\sim 50\%$; the null is an upper bound on a large effect, scoped to one capable-agent class and name-inferable tools.
- **Confounded axis.** Oracle-presence co-varies with headroom and task type; the one-axis thesis is suggested, not isolated.
- **ICC instability.** ICC swung 0.33 vs 0.57 across runs; effective- N (36.3) and the verdict's precision rest on a noisy estimate.
- **One corpus, one proposer.** $N=20$ on one repository's bug stream with one mini-class model; the dead-end pilots (metamorphic 0/8, held-out 3/8) are small- N with wide CIs.